

Organic matter degradation during weathering of Paleozoic black shales - a TG-MS study

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INTRODUCTION

Mineralization of dispersed organic matter (OM) from ancient black shales during weathering is one of the major factors controlling the atmospheric O_2 -level (Chang & Berner 2001). It has been shown that weathering causes significant changes in amount and composition of the residual organic matter in such rocks (e.g. Littke et al. 1991) and that biological processes play an important role in the transformation of ancient organic matter (Petsch et al. 2001). We investigated low-grade metamorphic black shales from the Silurian Graptolite Shale Series (Berga and Schwarzburg Anticline, Germany, Fig. 1). These shales show on several outcrops deep weathering profiles due to Mesozoic-Tertiary weathering. In the transition zone from bleached to unbleached shales weathering gradients can be observed inside single rock particles (Fig. 2). Aim of this study is to evaluate the changes of organic matter and minerals along these gradients in a mm-scale.



Fig. 1: Location of sampling

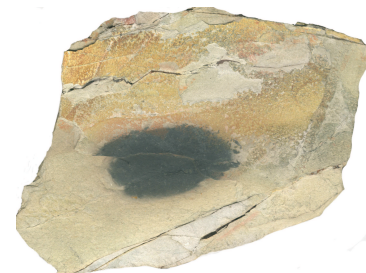


Fig. 2: Bleached alun shale specimen with black OM-rich core

METHOD AND SAMPLES

Combined TG/MS techniques have been used widely for the simultaneous detection of organic and inorganic volatiles in pyrolysis experiments. One major problem of these methods is the possibility of reverse reactions of evolved volatiles. To avoid this we used the pyrolysis technique DEGAS, consisting of a TG coupled directly to a quadrupole mass spectrometer (Fig. 4). Vacuum pyrolysis combined with direct mass spectrometric detection (no pressure reduction) has the advantage of highly nonequibrated conditions without the possibility of reverse reactions of the evolved volatiles. The system enables a temperature resolved analysis of low molecular weight fragments and the simultaneous detection of inorganic volatiles (e. g. H_2O , CO_2 , CO , N_2 , SO_2) during heating in vacuum (Schmidt & Heide 2001). For this study we used whole rock samples of Silurian black shales affected by recent (Fig. 3-1) and Mesozoic-tertiary weathering (Fig. 3-2/3).

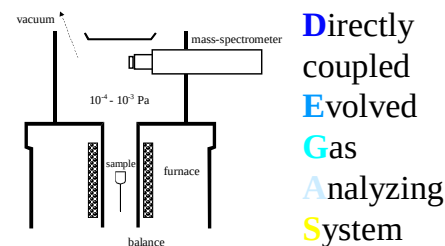


Fig. 4: DEGAS, directly coupled evolved gas analyzing system

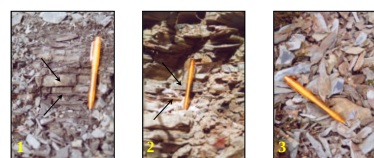


Fig. 3: Outcrop quarry „Weinbergbrüche“ Hohenleuben, Berga anticline

- 1 - interbedding of strongly and slightly silicified shales - recent weathering
- 2 - same formation - Mesozoic-Tertiary weathering
- 3 - bleached shale clasts with C_{org} -rich cores from the transition zone

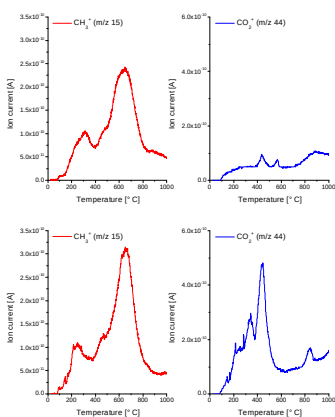


Fig. 5: Recent weathering after 50 years of exposure on a waste rock dump
- upper part: inner core
- lower part: outer rim of the specimen

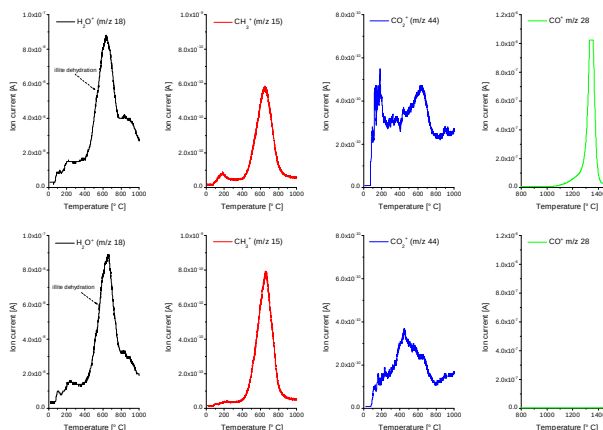


Fig. 6: Mesozoic-Tertiary weathering
- upper part: dark core
- lower part: bleached rim of the specimen
residual carbon is oxidized to CO at high temperatures

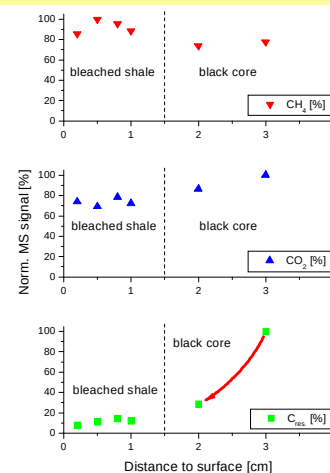


Fig. 7: Mesozoic-Tertiary weathering - normalized MS-signals as a function of distance to the surface

RESULTS AND CONCLUSIONS

Weathering of OM in low-grade metamorphic black shales during Mesozoic-Tertiary caused a drastic OM decrease, from approx. 23 wt. % down to approx. 1 wt. % (Fischer et al. 2003). Even white parts of the shales still contain about 1 wt. % OM. DEGAS-Analyses of weathering gradients across single specimens show a strong decrease in residual, non-pyrolisable carbon. (Fig. 7). This correlates with a decrease in pyrolysable CO_2 . Interestingly the amount of generated methane (detected as CH_3^+) remains nearly constant over the whole specimen. Temperature range and peak shape of methane release correlate strongly with the dehydration of illite (Fig. 6). We therefore assume an organo-illite complex as the source of most of the methane. From observations across weathering gradients we conclude, that this complex is extremely resistant against weathering. Comparison with data from elemental analysis shows, that OM from this complex forms up to 1 wt. % of the whole rock. From previous studies we know, that only kaolinisation of the illite in nature as well as HF-treatment of the samples in laboratory delete the found methane signal.

Samples affected by recent weathering (Fig. 5) show a decrease of CO_2 release from the outer part of the sample and a significant change in bitumen release temperature.